Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
BCA | Sem-3 | Summer-2024

Course Code	:	2104CS301	**Date**	:	15-05-2024
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Explain the differences between LAN, MAN, and WAN?	**4**
	(B)	Define Computer Network and list various applications where computer network is needed.	**3**
		OR	
		List common network topologies. Explain any one with proper figure.	
	(C)	Define circuit switching and packet switching. Explain the fundamental difference between them.	**7**
		OR	
		Describe a real-world scenario where circuit switching is preferred over packet switching.	
Q.2	**(A)**	List the four layers of the TCP/IP model and their corresponding protocols.	**4**
	(B)	Explain the function of the Physical layer in the OSI model.	**3**
		OR	
		Explain the function of the Presentation layer in the OSI model.	
	(C)	Explain the advantages and disadvantages of connection-oriented communication compared to connectionless communication.	**7**
		OR	
		Explain the function of network, and data link layer.	
Q.3	**(A)**	List and describe the types of transmission media used in networking.	**4**
	(B)	Draw the construction of coaxial cable in transmitting data signals.	**3**
		OR	
		Draw the construction of fiber optic cable in transmitting data signals.	
	(C)	Explain the advantages and limitations of infrared transmission compared to other wireless technologies.	**7**
		OR	
		Explain radio wave for wireless communication.	

Q.4	**(A)**	Define LAN card, and what function does it serve in a computer network?	**4**
	(B)	Draw a network infrastructure that uses repeaters to overcome signal degradation in large-scale deployments.	**3**
		OR	
		Draw a network infrastructure using hubs to interconnect devices in a small office environment.	
	(C)	Explain switch and router.	**7**
		OR	
		Explain bridge and gateway.	
Q.5	**(A)**	Explain the difference between IPv4 and IPv6 addressing schemes.	**4**
	(B)	Explain the concept of IP address classes and their respective ranges.	**3**
		OR	
		Describe the limitations of the classful addressing scheme. Define CIDR.	
	(C)	Define FTP and explain its role in transferring files over a network.	**7**
		OR	
		Define DNS and explain its role in translating domain names into IP addresses.	

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